23CSE202 DATABASE MANAGEMENT SYSTEMS

S3 B.Tech CSE

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PROJECT ABSTRACT

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Project Title : CONTACT MANAGEMENT SYSTEM

Abstract:

In today's digital environment, individuals interact with numerous contacts on a daily basis, including friends, family, colleagues, and clients. However, this information is frequently scattered across phone contact lists, email accounts, and physical records, with no unified structure to manage it. This often results in duplicate entries, incomplete details, and the absence of any record of past interactions. The proposed project, the Contact Management System (CMS), addresses this problem by designing a well structured relational database capable of storing, organizing, and managing contact information in a reliable and efficient manner.

The system is designed to represent the relationships between different types of data, rather than treating contact storage as a simple flat file. This includes handling scenarios such as a single contact belonging to multiple groups, tracking communication history over time, and enforcing different levels of access for administrators and regular users. The system is implemented using SQL (MySQL or PostgreSQL), with the database design forming the core focus of the project, while the user interface remains functional but secondary in scope.

The database consists of multiple interlinked tables. A Contacts table stores essential personal details, while a separate Groups table allows categorization such as family, work, or clients. Since a contact may belong to multiple groups and a group may contain multiple contacts, a junction table is used to represent this many to many relationship. A Communication Log table records calls, messages, and meetings associated with each contact through foreign key relationships, and a Users table manages role based access control, distinguishing the permissions of administrators from those of standard users.

Database complexity beyond simple read and write operations is achieved through triggers that detect duplicate contact entries prior to insertion, stored procedures that safely merge duplicate records without losing existing communication history, and views that generate summarized information such as frequently contacted individuals or group wise distribution of contacts. Indexing on frequently searched fields such as name and phone number is also incorporated to maintain query performance as the volume of data increases.

The intended users of this system include individuals seeking a structured approach to personal contact management, as well as small teams or organizations that require shared access with clear accountability. Administrators are able to oversee user permissions and communication records, while standard users manage their own contacts within defined limits. Overall, the project demonstrates meaningful database concepts, including relational integrity, constraints, and role based access, extending well beyond the scope of a basic contact storing application.